

- 4 In 2003, the compact mass spectrometer was one of the instruments sent by NASA to determine if Mars can support life by investigating compounds in the rocks and land such as carbon, hydrogen, oxygen, and nitrogen, and to determine their relative abundance.

Fig. 4.1 shows the schematic diagram of a compact mass spectrometer, which comprises a *velocity selector* with a magnetic flux density B of 0.200 T and an electric field strength E of 60 N C⁻¹, and a detector plate. A positive ion mixture is fed into the mass spectrometer, and the dotted line is the path taken by the positive ions that emerged from the velocity selector.

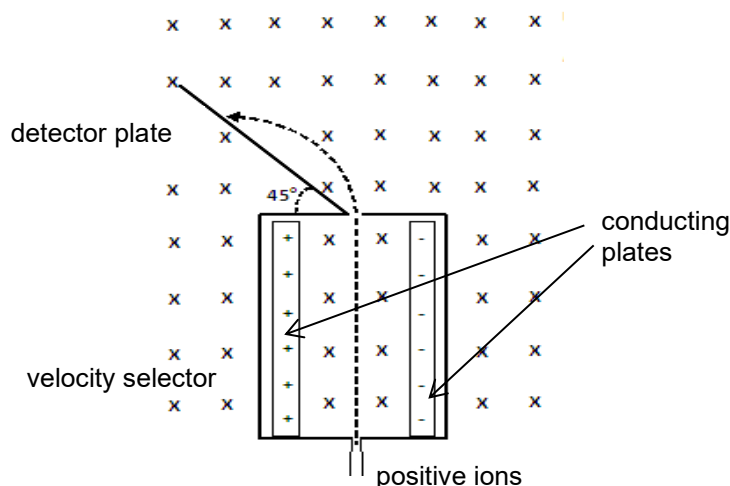


Fig 4.1

- (a) Positive ions each of charge q and mass m hit the detector plate.
- (i) By first determining the speed of the emergent ions from the velocity selector or otherwise, show that the radius of its subsequent circular path is

$$r = \frac{mE}{qB^2}$$

[3]

- (ii) Explain how the velocity of the positive ions that emerged from the velocity selector can be increased.

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 [1]

- (iii) The positive ions are now replaced with negative ions. Explain how the negative ions can still reach the detector plate as shown in Fig 4.1.

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 [2]

- (b) The detector plate then moved to a new horizontal position PQ as shown in Fig. 4.2. A rock sample vapour contains both $^{12}\text{C}^+$ and $^{13}\text{C}^+$ ions.



Fig 4.2

- (i) On Fig. 4.2, label the path for $^{13}\text{C}^+$ ions as **C**. [1]
- (ii) The radius of the circular paths for the two ions are 0.187 m and 0.203 m.

Determine the distance between the points of impact on the detector plate of the $^{12}\text{C}^+$ and $^{13}\text{C}^+$ ions.

distance = m [2]

[Turn over